



SCHOOL OF COMPUTING AND ENGINEERING SCIENCES
BACHELOR OF SCIENCE IN INFORMATICS AND COMPUTER SCIENCE
ICS 1204 INTEGRAL CALCULUS
Tutorial Exercise 2

Instructions: Answer all questions.

DATE: September 9, 2024

Part 1: Carry out the following integrations:

1. $\int 3 \sin^2 x \, dx = \frac{3}{2}x - \frac{3}{4} \sin 2x + c$
2. $\int (1 - \cos 2x)^2 \, dx = \frac{3}{2}x - \sin 2x + \frac{1}{8} \sin 4x + c$
3. $\int (2 + 2 \tan^2 x) \, dx = 2 \tan x + c$
4. $\int \frac{1 + \cos x}{\sin^2 x} \, dx = -\cot x - \operatorname{cosec} x + c$
5. $\int \frac{\cos^2 x}{1 + \sin x} \, dx = x + \cos x + c$
6. $\int \frac{\cos^2 x}{1 + \sin x} \, dx = x + \cos x + c$
7. $\int \sin x \sin 3x \, dx = \frac{1}{4} \sin 2x - \frac{1}{8} \sin 4x + c$
8. $\int \sin^2 2x \, dx = \frac{1}{2}x - \frac{1}{8} \sin 4x + c$
9. $\int 1 - \cot^2 x \, dx = 2x + \cot x + c$
10. $\int \frac{1}{\sec x - 1} \, dx = -x - \cot x - \operatorname{cosec} x + c$
11. $\int \frac{1}{\cos x \sin^2 x} \, dx = \ln |\sec x + \tan x| - \operatorname{cosec} x + c$
12. $\int \sin^2 x \sec^2 x \, dx = \tan x - x + c$
13. $\int \frac{\cos 2x}{\cos^2 x} \, dx = 2x - \tan x + c$
14. $\int \frac{1 - \cos x}{1 + \cos x} \, dx = 2 \tan(x/2) - x + c = -2 \cot x - x + 2 \operatorname{cosec} x + c$
15. $\int \sqrt{\sin^2 x + (\cos x - 1)^2} \, dx = -4 \cos(x/2) + c$

Part 2: Carry out the following integrations:

1. $\int_0^{\pi/2} 4 \sin^2 x \, dx = \pi$

2. $\int_0^{\pi/6} 8 \sin x \cos x \, dx = 1$

3. $\int_0^{\pi/6} (1 - \cos 3x)^2 \, dx = \frac{\pi}{4} - \frac{2}{3}$

4. $\int_{\pi/6}^{\pi/3} (3 \cot x + \tan x)^2 \, dx = \frac{2}{3}(10\sqrt{3} - \pi)$

5. $\int_0^{\pi/3} \frac{\sin x}{\cos^2 x} \, dx = 1$

6. $\int_0^{\pi/4} \tan^2 x \, dx = \frac{1}{4}(4 - \pi)$

7. $\int_0^{\pi/3} \frac{1}{1 - \sin x} \, dx = 1 + \sqrt{3}$

8. $\int_0^{\pi/2} (1 + \tan(x/2))^2 \, dx = 2 + \ln 4$

9. $\int_0^{\pi/2} \cos^3 x \, dx = \frac{2}{3}$

10. $\int_0^{\pi/12} 10 \sin 8\theta \cos 2\theta \, d\theta = \frac{1}{12}(16 + 3\sqrt{3})$

11. $\int_0^{\pi/4} (\cos x + \sec x)^2 \, dx = \frac{5}{8}(\pi + 2)$

12. $\int_0^{\pi/12} 6 \sin^2 \theta \, d\theta = \frac{1}{4}(\pi - 3)$

END
